

Gate 1 Record

GATE 1 — SOFTWARE AND LOGGING CHAIN VERIFICATION RECORD

CIP-BTSB-BT-001 · Supporting Record G1

CIP — Compute Infrastructure Platform · S. Mohammed Idreesh · Chennai Gate completed prior to the Gate 2 sensing check of 24 August 2026 · Record issued 25 August 2026

[PRE-GEN-0 MECHANISM DEMONSTRATION — NOT A PRODUCT VALIDATION]

1. PURPOSE

To establish, before any sensor or heater was connected, that the complete software chain — toolchain, board, library, sketch, and serial link — functions correctly and produces a machine-readable record at a known, verified sampling rate.

Gate 1 exists to separate software faults from hardware faults. Every subsequent gate assumes that when a reading looks wrong, the cause is physical. That assumption is only safe if the software was proven first, in isolation, against a target whose correct behaviour is known in advance.

This record is referenced by Section 4 of the pre-registered test report CIP-BTSB-BT-001 and is the foundation of Records G2 and G3.

2. CONFIGURATION AS BUILT

2.1 Toolchain and hardware

Item	Detail
Controller	Arduino Uno R3, ATmega328P, powered from laptop USB
Host	Windows laptop, Arduino IDE
Serial port	COM7, 9600 baud, 8-N-1
Library	“MAX6675 library” by Adafruit, v1.1.2
Sketch	BTSB_bench_logger.ino
Source size	2,543 bytes
Compiled size	4,882 bytes — 15% of the ATmega328P’s 32,256 byte flash
Flash headroom	85% free

The 85% headroom is recorded deliberately. It means the sketch can be extended during the test programme — additional channels, changed sampling interval, an added diagnostic — without any risk of running out of program space mid-campaign.

2.2 Pin map as compiled

```
const int SCK_PIN = 6;    // shared clock, both modules
const int SO_PIN = 5;     // shared data, both modules
const int CS1_PIN = 7;    // chip select – TC1, HOT plate
const int CS2_PIN = 8;    // chip select – TC2, COLD plate

const float T_LIMIT = 120.0; // deg C on TC1
```

Two MAX6675 amplifiers share the clock and data lines and are distinguished only by their chip-select pins. **Module #1 on D7 is TC1, the hot plate. Module #2 on D8 is TC2, the cold plate.** This assignment is fixed in firmware and must not be changed between runs.

2.3 Output format

```
# BTSB bench test - dual thermocouple log
# TC1 = HOT plate, TC2 = COLD plate, temps in deg C
# Hard limit 120 C on TC1 - cut power if ALARM appears
time_s,T1_hot_C,T2_cold_C,delta_C
0.3,33.75,33.75,0.00
```

Four columns, comma-separated, two decimal places, one line per second, with a three-line commented header. The format is directly loadable by the analysis tool without transcription — which removes an entire class of transcription error from the programme.

3. METHOD

Verification proceeded in two stages, deliberately ordered.

Stage 1 — Blink. The stock `Blink` example was compiled and uploaded before any project code was touched. Blink exercises the whole chain — IDE, compiler, USB driver, bootloader, board — against a target whose correct behaviour is visible across the room and requires no instrumentation to judge.

Stage 2 — Logger. `BTSB_bench_logger.ino` was compiled, uploaded, and its serial output observed with **no MAX6675 modules connected at all**.

Testing the logger with nothing attached is the point of the exercise. It establishes what the system prints when the sensing hardware is absent, so that this signature can be distinguished from a genuine sensor fault later on.

4. RESULTS

4.1 Toolchain

Check	Result
Board detected on COM7	Pass
Adafruit MAX6675 library installed	Pass
<code>BTSB_bench_logger.ino</code> compiles clean	Pass
Upload completes, TX/RX activity observed	Pass
Serial Monitor prints the CSV header	Pass

4.2 Sampling interval — the governing value

The loop is constructed as:

```
readCelsius(TC1) → delay(250) → readCelsius(TC2) → delay(750)
```

The 250 ms delay is not padding. The MAX6675 requires approximately 220 ms to complete a conversion, and reading the second channel sooner would return a stale or corrupt value.

Nominal interval is 1.000 s. Measured against the Gate 2 dataset:

```
836 readings spanning 0.3 s → 837.5 s
mean interval = 837.2 / 835 = 1.0026 s
```

Measured sampling interval: 1.003 s, i.e. 0.26% above nominal.

This matters. The analysis integrates temperature against time to obtain energy, and compares intervals between runs. A timebase error propagates directly into the energy balance. At 0.26% it is two orders of magnitude below the other uncertainties in the experiment and can be neglected — but it is neglected because it was measured, not because it was assumed.

4.3 Behaviour with no sensor connected

With no MAX6675 modules present, the system printed:

```
0.3,0.00,0.00,0.00
```

Not NaN. See Section 6.2 — this result contradicted the prediction given to the operator, and the discrepancy proved to have diagnostic value.

4.4 Fault detection path

```
if (isnan(T1) || isnan(T2)) {
  Serial.println(",NaN,NaN,NaN <<< SENSOR FAULT - check probe wiring");
}
```

The `NaN` path is asserted only when a module is present and reports an open thermocouple. Its correct operation was confirmed later, during Gate 2, where it successfully flagged an intermittent open circuit at the probe termination — a fault that would otherwise have appeared as silently corrupted data.

4.5 Safety alarm — and its explicit limitation

```
if (T1 >= T_LIMIT) {
  Serial.print(" *** ALARM: TC1 OVER 120 C - CUT POWER NOW ***");
}
```

The sketch cannot cut power. It prints a warning. The operator cuts power.

This is stated in the sketch’s own header comment and is repeated here because it is a safety-critical limitation, not an implementation detail. The 120 °C limit is advisory only.

The physical abort mechanism is the **rocker switch in the positive supply line**, documented in Record G3, positioned within the operator’s reach. Software raises the alarm; the operator’s hand is the interlock. Any account of this test programme that implies automatic thermal cut-out would be false.

5. INTERPRETATION

Quantity	Value	Basis
Sampling interval	1.003 s	measured over 836 samples
Timebase error	+0.26%	measured
Flash utilisation	15%	compiler report
Output format	4-column CSV, 1 Hz	verified
Fault detection	present, verified in Gate 2	verified
Thermal cut-out	none — advisory alarm only	by design

The logging chain is fit for the pre-registered analysis. Timestamps may be used directly for interval and rate calculations without correction.

6. FAULTS FOUND AND CORRECTED

Recorded in full because they form part of the instrument’s history.

6.1 Blink verification — asymmetric flash rate

Symptom. With both `delay()` values expected to have been changed to 200 ms, the LED flashed unevenly rather than at a uniform fast rate.

Cause. Only one of the two delay calls had been edited — line 34 was set to 200, line 36 remained at 1000. The result was a short on-period followed by a long off-period.

Correction. Both delay values aligned. Uniform flash observed.

Secondary error, mine. The rate predicted to the operator for two 200 ms delays was “5 blinks per second”. This is arithmetically wrong: two 200 ms delays give a 400 ms period, i.e. **2.5 blinks per second**. Corrected on discovery.

6.2 Predicted NaN, observed 0.00 — a corrected assumption

Prediction given to the operator. With no MAX6675 connected, the logger would print NaN and trigger the sensor-fault line.

Observed. 0.00.

Cause. With no module attached, the shared SO data line is not driven. The library clocks in a run of zeros, which decodes as a valid reading of 0 °C. The `isnan` fault path is never reached, because nothing reported a fault — nothing reported anything at all.

Why this matters. The discrepancy is more useful than the prediction would have been. It yields a three-way diagnostic that was used repeatedly in Gate 2:

Serial Monitor shows	Meaning
0.00	Module not connected — no data line activity
NaN	Module connected, probe open — check the screw terminal
~ambient, e.g. 33.75	Working correctly

Without this correction, a disconnected module would have been indistinguishable from a working one reading a cold surface.

6.3 Serial port busy

Symptom. Upload failed with a port-access error.

Cause. A second Arduino IDE window was open and holding COM7. Only one process may own a serial port.

Correction. All IDE windows closed, USB unplugged and replugged, a single window reopened.

6.4 Serial Monitor pane collapsed

Symptom. Serial Monitor appeared to produce no output.

Cause. The Monitor pane had been dragged to zero height within the IDE window. Output was being produced normally but was not visible.

Correction. Divider dragged upward.

7. OUTSTANDING

☒ Nothing outstanding for Gate 1.

The sketch is resident in Arduino flash and must NOT be re-uploaded before the test runs. Re-uploading resets the millis() timebase and would invalidate any run in progress.

8. GATE STATUS

GATE 1 — PASSED.

Toolchain, board, library, sketch, serial link and CSV output all verified. Sampling interval measured at 1.003 s. Fault-detection path present and subsequently proven in Gate 2.

No provisional conditions.

Record G1 · Issued 25 August 2026 · Retrospective, compiled from build-log entries and the Gate 2 dataset. Supports CIP-BTSB-BT-001. All build-manual performance figures remain [GEN-0 VALIDATION REQUIRED].